

Online Supplement to “Market Integration, Platformization, and Local Multipliers”

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A Proofs and Demand Details

Proof of Lemma 1. Write $\gamma = 1 - \alpha - \beta > 0$. The first-order conditions for the outer household problem are

$$\frac{\alpha}{M} = \lambda P_M, \quad \frac{\beta}{N} = \lambda, \quad \frac{\gamma}{Z} = \lambda.$$

The budget constraint gives $\lambda = 1/Y$, hence expenditures on (M, N, Z) equal $(\alpha Y, \beta Y, \gamma Y)$. Substitution into utility gives

$$R = K \frac{Y}{P_M^\alpha}, \quad K = \alpha^\alpha \beta^\beta \gamma^\gamma. \quad (\text{A.1})$$

Let $h_P(p_A, p_P, 1)$ be the platform Hicksian demand for one unit of the homothetic retail aggregate. Differentiability of the unit-expenditure function and Shephard’s lemma imply

$$\frac{\partial P_M}{\partial p_P} = h_P(p_A, p_P, 1), \quad s = \frac{p_P h_P(p_A, p_P, 1)}{P_M} = \frac{\partial \ln P_M}{\partial \ln p_P}.$$

Because p_A is fixed and $d \ln p_P / dz = -1$, the chain rule yields $d \ln P_M / dz = -s$, proving the demand identities in the lemma. $\square$

Proof of Lemma 2. For fixed (z, κ, G) , define

$$T(Y) = B + \omega G + (\beta + \alpha \omega)Y.$$

Since $s \in (0, 1)$ and both retention shares lie in $[0, 1]$, $0 \leq \omega \leq 1$. Thus T maps $\mathbb{R}_+$ into itself and

$$|T(Y) - T(\tilde{Y})| = (\beta + \alpha \omega)|Y - \tilde{Y}|, \quad \beta + \alpha \omega \leq \alpha + \beta < 1.$$

The contraction theorem gives a unique fixed point and convergence of iteration from every non-negative initial value. Solving the fixed-point equation yields (7). Its numerator is positive, while $D \geq 1 - \alpha - \beta > 0$, so $Y > 0$. Lemma 1 then determines the remaining household objects uniquely. $\square$

A.1 Benchmark multipliers

Holding (z, κ, G) fixed and varying the realized autonomous-income injection or outside-funded retail spending gives

$$\frac{\partial Y}{\partial B} = \frac{1}{D}, \quad \frac{\partial Y}{\partial G} = \frac{\omega}{D} = \omega \sum_{n=0}^{\infty} (\beta + \alpha \omega)^n.$$

The series converges because $\beta + \alpha \omega < 1$. It records that retail leakage recurs in every induced spending round.

Proof of Proposition 1. The ratio identities follow immediately from $Y = B/D$ and $R = KY/P_M^\alpha$. Organizational capacity does not affect B_j , s_j , or P_{Mj} . Differentiating

$$\mathcal{D}(\kappa) = \frac{D_1(\kappa)}{D_0(\kappa)}$$

therefore gives

$$\begin{aligned} \frac{\partial \ln \mathcal{D}(\kappa)}{\partial \kappa} &= -\alpha \delta_\rho \left[\frac{s_1}{D_1(\kappa)} - \frac{s_0}{D_0(\kappa)} \right] \\ &= -\frac{\alpha \delta_\rho d_A (s_1 - s_0)}{D_1(\kappa) D_0(\kappa)} < 0. \end{aligned} \quad (\text{A.2})$$

Hence both finite income ratios are strictly increasing in κ .

For any $q \in (\mathcal{D}(1), \mathcal{D}(0))$, continuity and strict monotonicity give a unique $\kappa(q) \in (0, 1)$ satisfying $\mathcal{D}(\kappa(q)) = q$. Solving

$$d_A + \alpha s_1 \Delta(\kappa) = q[d_A + \alpha s_0 \Delta(\kappa)]$$

gives (12). The denominator in that expression is positive at a solution because

$$s_1 - q s_0 = \frac{s_1 D_0(\kappa) - s_0 D_1(\kappa)}{D_0(\kappa)} = \frac{d_A (s_1 - s_0)}{D_0(\kappa)} > 0.$$

Since $0 < \mathcal{P} < 1$, $q_R^F = \mathcal{B}/\mathcal{P}^\alpha > q_Y^F = \mathcal{B}$. The inverse of the strictly decreasing function $\mathcal{D}$ is decreasing, so $\kappa_R^F < \kappa_Y^F$. Comparing $\mathcal{D}(\kappa)$ with the two targets proves the three strict regions. At the first threshold,

$$\frac{R_1}{R_0} = 1, \quad \frac{Y_1}{Y_0} = \mathcal{P}^\alpha < 1;$$

at the second,

$$\frac{Y_1}{Y_0} = 1, \quad \frac{R_1}{R_0} = \mathcal{P}^{-\alpha} > 1.$$

□

Proof of Corollary 1. At $G = 0$,

$$Y = \frac{B}{D}, \quad D = d_A + \alpha s \Delta(\kappa), \quad D_\kappa = -\alpha s \delta_\rho.$$

Therefore $\partial \ln Y / \partial \kappa = \alpha s \delta_\rho / D$. Since $D_z = \alpha \Delta(\kappa) s_z$,

$$\frac{\partial^2 \ln Y}{\partial z \partial \kappa} = \alpha \delta_\rho \frac{s_z D - s D_z}{D^2} \quad (\text{A.3})$$

$$= \frac{\alpha \delta_\rho d_A s_z}{D^2} > 0. \quad (\text{A.4})$$

Finally, $\ln R = \ln K + \ln Y - \alpha \ln P_M$, and P_M is independent of κ , so the same cross-partial holds for $\ln R$. □

Proof of Proposition 2. Strict concavity, the endpoint restrictions, and interior positivity imply $g'(0) > 0$ and $g'(1) < 0$. For example, for every $u \in (0, 1)$,

$$g'(0) \geq \frac{g(u) - g(0)}{u} > 0, \quad g'(1) \leq \frac{g(1) - g(u)}{1 - u} < 0.$$

Write $b = \alpha\Delta(\kappa) > 0$ and define

$$F(s, b) = g'(s)(d_A + bs) - bg(s).$$

Direct differentiation gives

$$\frac{\partial \Lambda_g}{\partial s} = \frac{bF(s, b)}{(d_A + bs)^2}, \quad F_s(s, b) = g''(s)(d_A + bs) < 0. \quad (\text{A.5})$$

Moreover,

$$F(0, b) = d_A g'(0) > 0, \quad F(1, b) = (d_A + b)g'(1) < 0.$$

Thus F has exactly one zero $s^*(b) \in (0, 1)$. Equation (A.5) shows that Λ_g rises before this zero and falls after it. Since its endpoint values are zero and it is positive in the interior, s^* is the unique global maximizer.

If $g(s) = g(1 - s)$, differentiability gives $g'(1/2) = 0$, so $F(1/2, b) = -bg(1/2) < 0$. Because $F(0, b) > 0$ and F is strictly decreasing, $s^*(b) < 1/2$.

For the capacity comparative statics, evaluate expressions at $s = s^*(b)$ and write $D^* = d_A + bs^*$. The first-order condition implies $g'(s^*) = bg(s^*)/D^*$, and hence

$$F_b(s^*, b) = s^*g'(s^*) - g(s^*) = -\frac{d_A g(s^*)}{D^*} < 0.$$

The implicit-function theorem, $b_\kappa = -\alpha\delta_\rho$, and the envelope theorem yield

$$\frac{\partial s^*}{\partial \kappa} = -\frac{\alpha\delta_\rho d_A g(s^*)}{g''(s^*)[D^*]^2} > 0, \quad (\text{A.6})$$

$$\frac{\partial \Lambda_g^{\max}}{\partial \kappa} = -\frac{\alpha\delta_\rho d_A g(s^*)}{[D^*]^2} < 0. \quad (\text{A.7})$$

□

A.2 Admissibility of the general share-speed

The share-speed restriction is consistent with a two-channel homogeneous demand system. Let $t = \ln(p_P/p_A)$ and take any interior solution of $ds/dt = -g(s)$. Define

$$\phi(r) = \exp \left\{ \int_{r_0}^r s(\ln u) d \ln u \right\}, \quad P_M(p_A, p_P) = p_A \phi(p_P/p_A).$$

Then P_M is linearly homogeneous and its platform expenditure share is $p_P P_{M, p_P} / P_M = s$. It is increasing because the two expenditure shares are s and $1 - s$. Moreover,

$$\phi''(r) = -\frac{\phi(r)}{r^2} [g(s) + s(1 - s)] < 0,$$

so P_M is a valid concave unit-expenditure function. This construction also shows that the non-CES share-speed in Figure 1 is not merely an arbitrary curve detached from a demand system.

A.3 CES specialization

The main assumptions nest the two-channel CES aggregator

$$M = \left[(1-a)^{1/\eta} M_A^{(\eta-1)/\eta} + a^{1/\eta} M_P^{(\eta-1)/\eta} \right]^{\eta/(\eta-1)}, \quad a \in (0,1), \quad \eta > 1. \quad (\text{A.8})$$

Cost minimization gives

$$P_M(z) = \left[(1-a)p_A^{1-\eta} + ap_P(z)^{1-\eta} \right]^{1/(1-\eta)}, \quad (\text{A.9})$$

$$s(z) = \frac{ap_P(z)^{1-\eta}}{(1-a)p_A^{1-\eta} + ap_P(z)^{1-\eta}}, \quad (\text{A.10})$$

$$s'(z) = (\eta-1)s(1-s). \quad (\text{A.11})$$

Thus $g(s) = (\eta-1)s(1-s)$ satisfies all conditions in Proposition 2, including symmetry. With $t(\kappa) = \sqrt{1 + \alpha\Delta(\kappa)/d_A}$, its unique peak and height have the closed forms

$$s^*(\kappa) = \frac{1}{1+t(\kappa)} < \frac{1}{2}, \quad (\text{A.12})$$

$$\Lambda_g^{\max}(\kappa) = (\eta-1) \frac{t(\kappa) - 1}{t(\kappa) + 1}. \quad (\text{A.13})$$

These expressions are the CES specialization, not assumptions needed for the general peak result.

As $\eta \downarrow 1$, the CES system approaches the Cobb–Douglas channel case, $P_M \rightarrow p_A^{1-a} p_P^a$ and $s \rightarrow a$. At fixed unequal prices, $\eta \rightarrow \infty$ concentrates expenditure on the cheaper channel. These are limits of the specialization rather than finite-parameter corners in the main interior demand path.

A.4 Boundary cases and outside-funded spending

If $\Delta(\kappa) = 0$, then $\Lambda_g \equiv 0$, so no unique drag peak exists. If $\Delta(\kappa) < 0$, integration shifts expenditure toward the higher-retention channel and the capture-drag proposition does not apply. The capacity derivatives in Proposition 2 are ordinary interior derivatives; feasible one-sided versions apply at $\kappa = 0, 1$. Without symmetry of g , the unique peak remains but need not lie below one-half.

The endpoint-crossing condition in Proposition 1 is sufficient for two interior capacity thresholds. A target strictly outside $[\mathcal{D}(1), \mathcal{D}(0)]$ has no threshold on $[0, 1]$; equality with an endpoint places the threshold at $\kappa = 1$ or 0 , respectively. Only a target in $(\mathcal{D}(1), \mathcal{D}(0))$ yields an interior threshold. The finite ratio identities remain valid in every case.

For completeness, when nominal outside-funded spending $G > 0$ is fixed,

$$\frac{d \ln Y}{dz} = \frac{\varepsilon B + \omega_z G}{B + \omega G} + \frac{\alpha \omega_z}{D}, \quad (\text{A.14})$$

$$\frac{d \ln R}{dz} = \frac{\varepsilon B + \omega_z G}{B + \omega G} + \frac{\alpha \omega_z}{D} + \alpha s, \quad (\text{A.15})$$

$$\frac{\partial \ln Y}{\partial \kappa} = s \delta_\rho \left[\frac{G}{B + \omega G} + \frac{\alpha}{D} \right] > 0, \quad \frac{\partial \ln R}{\partial \kappa} = \frac{\partial \ln Y}{\partial \kappa}. \quad (\text{A.16})$$

For a finite change, define

$$\mathcal{N} = \frac{B_1 + \omega_1 G}{B_0 + \omega_0 G}.$$

Then $Y_1/Y_0 = \mathcal{N}/\mathcal{D}$ and $R_1/R_0 = \mathcal{N}/(\mathcal{D}\mathcal{P}^\alpha)$. Because $\mathcal{N}$ itself varies with capacity, the closed-form M1 capacity thresholds are stated for $G = 0$.